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Sorghu-Shield: Eco Food Packaging

BATCH NO :-1

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Introduction :-

CONTEXT FOR ECO PACKAGING

The **growing plastic pollution problem** necessitates sustainable solutions. Eco-friendly packaging is vital for the food industry to reduce environmental impact.

Problem Statement - Plastic causes pollution - Existing packaging fails with hot food - Need eco-friendly solution

Solution / Idea - Use sweet sorghum bagasse - Biodegradable, strong, leak-proof - Replaces plastic containers

Challenges with Current Packaging

PLASTIC

Plastic materials cause severe **environmental damage**, contributing to pollution and harming wildlife in ecosystems around the world.

PAPER

Traditional paper products often **leak** when used with hot food, limiting their usability and effectiveness in the food industry.

ALTERNATIVES

There is a significant **lack of strong**, biodegradable alternatives in the market, creating a need for innovative packaging solutions.

Introducing Sorghu-Shield: Sustainable Packaging Solution

ECO-FRIENDLY AND BIODEGRADABLE

Sorghu-Shield utilizes **sweet sorghum bagasse** to create strong, leak-proof containers, offering a sustainable alternative to conventional plastic packaging.

Unique Features

INNOVATION

Sorghu-Shield represents **novel innovation** in eco-packaging through the use of sustainable sorghum fiber, reducing reliance on traditional materials.

TECHNOLOGY

The application of **high-temperature hot pressing technology** enhances the strength and durability of packaging, ensuring it meets food safety standards.

BIODEGRADABILITY

Fully **biodegradable and compostable**, Sorghu-Shield packaging contributes to a sustainable waste management system, supporting environmental goals effectively.

Sweet Sorghum Stalk

STRONG AND SUSTAINABLE RAW MATERIAL SOURCE

The outer stalk of sweet sorghum is a **robust fiber source** ideal for eco-friendly packaging. With a **growth cycle** of 120–135 days, it requires moderate water supply, making it suitable for regions like Andhra Pradesh, where its availability aligns with local agricultural practices.

Manufacturing Process

MATERIAL SOURCING

Sourcing sweet sorghum for sustainable production.

CLEANING PROCESS

Thoroughly cleaning sorghum to ensure quality.

PACKAGING FORMATION

Molding and shaping into final product containers.

Key Manufacturing Steps of Sorghu-Shield

UNDERSTANDING THE PRODUCTION PROCESS

The manufacturing process includes pulping, forming, hot pressing, and sterilization, ensuring that our containers remain safe and durable for all food types.

Pulping: fiber + water mix
- Forming: shape creation
- Hot pressing: strength
- Sterilization: food safety

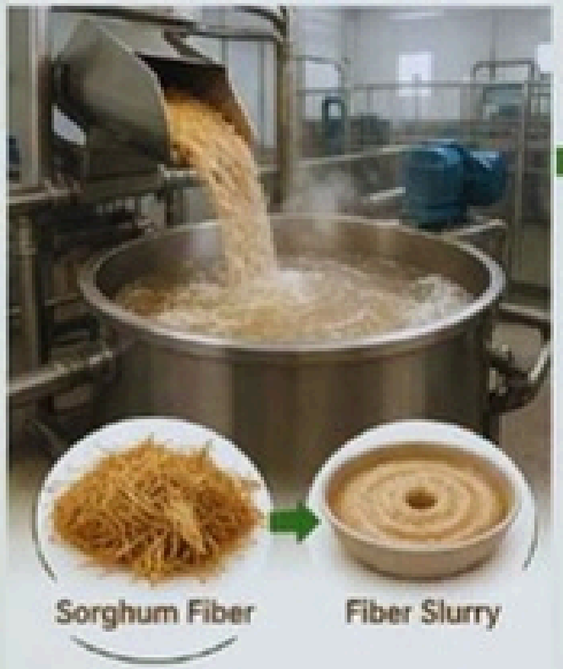
1. Raw Material Collection
↓
2. Cleaning & Drying
↓
3. Crushing / Size Reduction
↓
4. Pulping (Fiber + Water Mixing)
↓
5. Screening & Refining
↓
6. Molding / Forming (Shape Creation)
↓
7. Vacuum Dewatering
↓
8. Hot Pressing (High Temperature & Pressure)
↓
9. Trimming & Finishing
↓
10. Sterilization (Heat / UV Treatment)
↓
11. Quality Checking

MACHINERY USED IN PRODUCTION

Turning Sorghum Fiber into Eco-Friendly Food Packaging

1. PULPING

Fiber into Slurry



- ✓ Fibers mixed with water & chemicals
- ✓ Converted into smooth pulp

HYDRAPULPER
(Pulper Machine)

2. FORMING

Shape Creation

- ✓ Slurry poured into molds
- ✓ Vacuum removes water

VACUUM FORMING MACHINE
(Molding Machine)

3. HOT PRESSING

Strength & Smooth Finish

- ✓ Trays compressed with heat & pressure
- ✓ Stronger, durable & smooth

HOT PRESS MACHINE
(Thermo-press)

. STERILIZATION

Food Safety

- ✓ UV light kills bacteria & germs
- ✓ Ready for food packaging

UV STERILIZATION UNIT
(Conveyor UV Chamber)

Advantages of Sorghu-Shield

ECO-FRIENDLY

Sorghu-Shield is made from **renewable materials**, ensuring a minimal environmental footprint and supporting sustainable practices in food packaging.

LEAK-PROOF

Designed to be **leak-proof**, Sorghu-Shield containers can hold hot, liquid foods without compromising integrity, ensuring customer satisfaction.

COST-EFFECTIVE

With a production cost of **₹1.90 per box**, Sorghu-Shield offers a budget-friendly alternative to traditional packaging options, enhancing profitability.

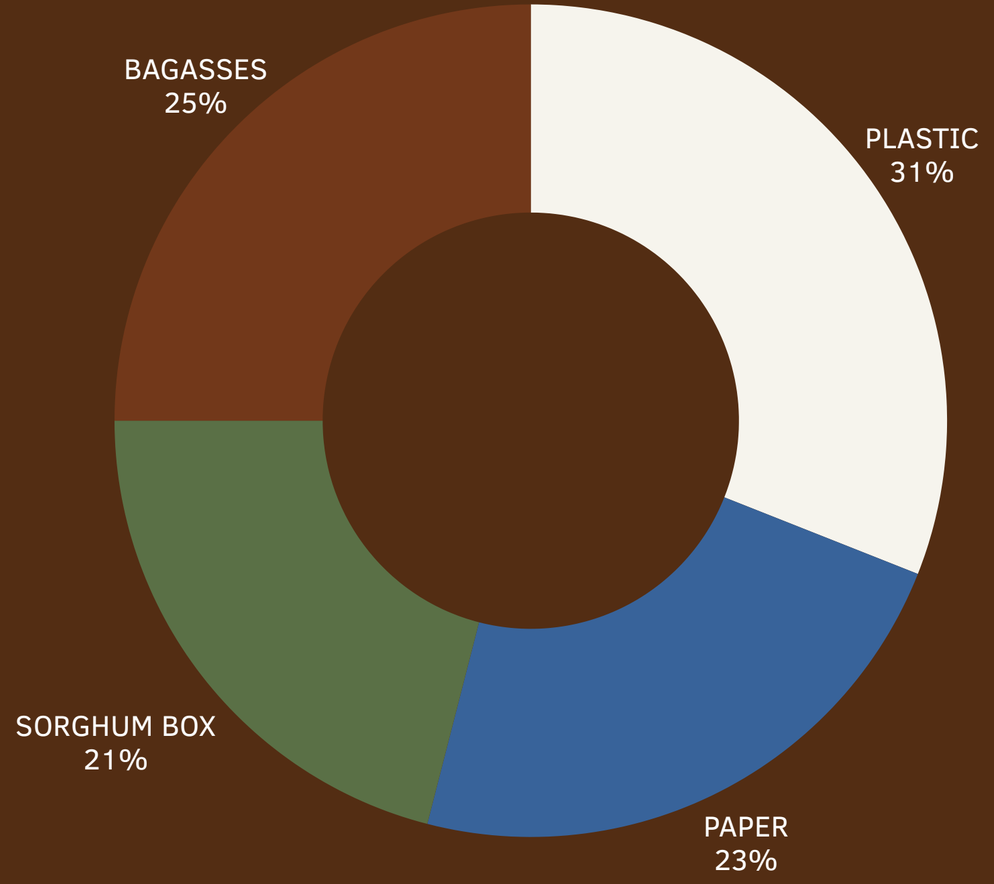
Comparing Sorghum Packaging with Alternatives

MATERIAL STRENGTH AND ENVIRONMENTAL IMPACT

Sorghu-Shield packaging outperforms traditional materials like plastic and paper in strength, leak-proof capabilities, and eco-friendliness while being cost-effective.

Cost Comparison of Eco Packaging

The chart illustrates the **cost-effectiveness** of Sorghu-Shield packaging compared to alternatives. Sorghu-Shield emerges as the **most affordable** option, promoting sustainability without compromising quality.



Cost Comparison of Sorghum Packaging Against Alternatives

Waste Management Insights

WASTE GENERATION

Our process generates only **8–15% waste**, making it an efficient option for eco-friendly production without compromising on quality.

RECYCLING RATE

With a **90–95% recycling rate**, we ensure that most of our materials are repurposed, significantly reducing environmental impact.

WATER REUSE

We implement a **water reuse strategy**, allowing us to recycle water within the production process, conserving this vital resource effectively.

₹1.10

PROFIT PER BOX

The **profit per box** for Sorghu-Shield packaging is ₹1.10. This indicates a **strong financial** performance, highlighting the potential for sustainable packaging solutions to be both eco-friendly and economically viable. As the demand for green packaging rises, profitability is expected to improve even further.

Selling price: ₹3.00
Profit: ₹1.10
Profit margin: 36–37%

Production and Break-Even

VOLUME

We plan to produce **50,000 boxes** per month, meeting the rising demand for eco-friendly packaging in the food industry.

PROFIT

The **monthly profit** is projected to reach ₹55,000, demonstrating the financial viability and sustainability of our product.

TIMELINE

The estimated **break-even point** is approximately 14 months, making Sorghu-Shield a promising investment opportunity for stakeholders.

Future Potential of Eco-Friendly Packaging Solutions

TRENDS IN MARKET GROWTH

The demand for **sustainable packaging** solutions is rising significantly, driven by consumer awareness and government initiatives against plastic pollution. Increasing demand for eco-products - Government plastic ban support - Expansion to food industry **Conclusion**: - Sustainable alternative to plastic - Profitable business model - Future-ready innovation